

Predicting Fraudulent Banking Transactions Using AI: A Decision-Support Approach for Settlement Risk Management

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1. Introduction

Fraud in banking and payment settlement systems represents a persistent and evolving risk for financial institutions. As digital payment volumes increase and settlement processes accelerate, banks are required to manage fraud risk while maintaining operational efficiency, regulatory compliance, and customer trust.

Fraud has significant consequences for banks, including direct financial losses, increased operational workload for fraud and compliance teams, and erosion of customer trust. Manual investigation processes can become overwhelmed as transaction volumes grow, while overly rigid rule-based systems often generate high numbers of false positives, delaying legitimate settlements and increasing operational friction.

In response to these challenges, artificial intelligence has increasingly been explored as a tool to support fraud risk management. Rather than fully automating decisions, AI can be used to assist fraud analysts by identifying high-risk transactions and prioritising investigations. This decision-support approach enables banks to manage risk more effectively while maintaining human oversight, transparency, and accountability within settlement workflows.

2. Problem Statement

Fraud detection in banking environments is particularly challenging due to the scale and complexity of modern transaction systems. Financial institutions must monitor extremely large volumes of transactions in near real time, while fraud patterns continuously evolve in response to new controls

and regulatory measures. As a result, traditional fraud detection approaches often struggle to adapt effectively to changing risk conditions.

In practice, many banks rely on rule-based fraud detection systems that are designed to flag potentially suspicious transactions. While these systems can be effective in identifying certain types of fraud, they often generate high numbers of false positives. This leads to unnecessary manual investigations, increased workload for fraud analysts, and delays in settlement for legitimate customer transactions.

The objective of this project is to propose an AI-based decision-support approach that assigns fraud risk scores to individual transactions. Rather than automatically blocking transactions, the system is intended to support fraud analysts by prioritising cases for investigation within settlement workflows, enabling more effective risk management while maintaining human oversight.

3. Context: Irish & EU Banking Environment

Banks operating within Ireland and the wider European Union are subject to strict regulatory requirements related to data protection, financial crime prevention, and customer rights. Regulations such as the General Data Protection Regulation (GDPR) and anti-money laundering frameworks impose obligations on financial institutions to ensure transparency, accountability, and fairness in decision-making processes.

Within this regulatory environment, the use of fully automated decision-making systems presents significant challenges. Under GDPR, individuals have rights related to decisions that are made solely through automated processing, particularly where such decisions may have legal or significant effects. In banking contexts, this means that fraud-related decisions must be explainable, auditable, and subject to appropriate human oversight.

In light of these regulatory and governance constraints, this project adopts a decision-support approach rather than a fully automated fraud prevention system. The proposed use of AI is intended to assist human analysts by highlighting transactions with elevated fraud risk, while final decisions remain under human control. This approach supports compliance requirements and enables organisations to manage fraud risk responsibly during periods of technological and operational change.

4. Dataset Description

This study uses a publicly available financial transactions dataset commonly employed in academic research for fraud detection and risk analysis. The dataset contains simulated banking transaction records that reflect real-world payment and settlement behaviour, including transaction amounts, transaction types, account balances, and fraud indicators.

Each record represents a single transaction and includes attributes related to the origin and destination accounts, transaction value, and transactional context. The target variable, *isFraud*, indicates whether a transaction was identified as fraudulent (1) or legitimate (0). This binary classification setup aligns with typical fraud risk assessment workflows in banking environments.

Dataset Overview

Attribute	Description
Dataset	Financial Transactions Fraud Dataset
Records	Large-scale transactional dataset (millions of records)
Target Variable	isFraud (Fraudulent = 1, Legitimate = 0)
Key Features	Transaction amount, transaction type, account balances, transaction flow

The dataset was selected due to its relevance to payment settlement systems and its suitability for demonstrating risk-based fraud analysis. Although the data is simulated, it reflects realistic transaction patterns and fraud behaviours observed in banking operations, making it appropriate for educational and decision-support modelling purposes.

Class Imbalance in Fraud Detection Data

Exploratory analysis revealed a severe class imbalance within the target variable (*isFraud*). Fraudulent transactions account for approximately **0.12%** of the total records, while **99.88%** represent legitimate transactions. This imbalance mirrors real-world banking environments, where fraudulent events are rare but carry significant financial and operational consequences.

Such imbalance poses a major challenge for fraud detection systems. Models optimised solely for accuracy may appear effective by predicting the majority non-fraud class, yet fail to detect fraudulent transactions. In settlement and payments workflows, this can lead to undetected fraud or excessive false positives, increasing analyst workload and negatively impacting customer trust.

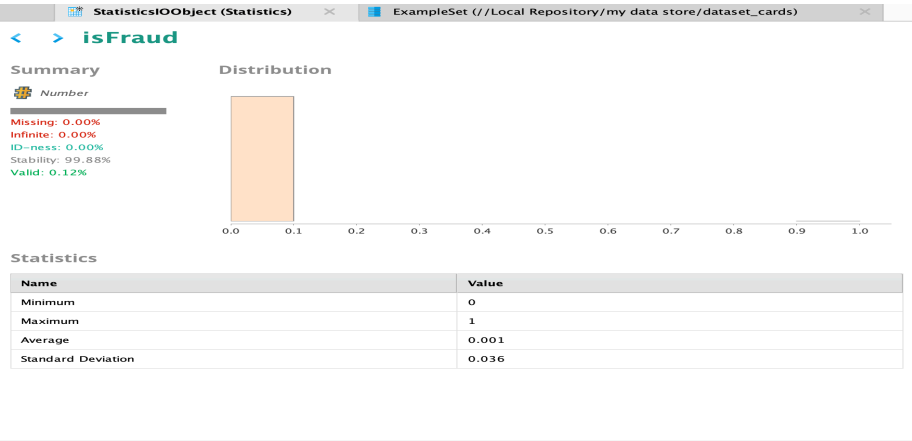


Figure X: Distribution of Fraudulent vs Non-Fraudulent Transactions (isFraud). The dataset exhibits severe class imbalance, with fraud cases representing

Accordingly, this project adopts a **decision-support approach**, where the objective is not automated fraud blocking but the identification of higher-risk transactions for human review. This aligns with regulatory expectations within the Irish and EU banking context, particularly under GDPR and AML frameworks, which emphasise explainability, proportionality, and human oversight.

Transaction Amount Characteristics

Analysis of the transaction amount feature highlights a wide range of values, with a small number of extremely high-value transactions and a large concentration of lower-value payments. The distribution is heavily right-skewed, indicating the presence of outliers that may represent elevated settlement risk.

From an operational perspective, transaction amount is a critical indicator for fraud prioritisation. High-value transactions, even when infrequent, may warrant additional scrutiny due to their potential financial impact. This reinforces the need for risk scoring rather than binary automated decisions, enabling analysts to focus attention where potential losses are greatest.

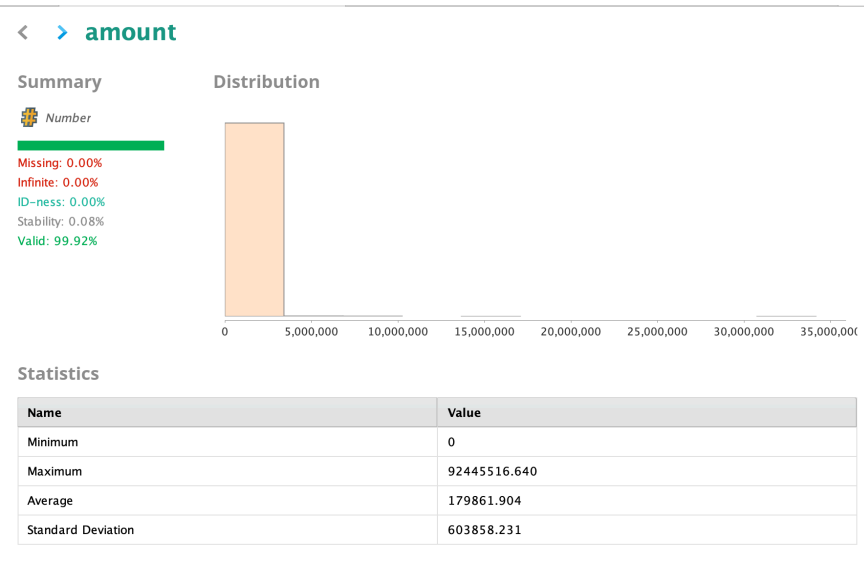


Figure Y: Distribution of Transaction Amounts. The data shows a wide range of transaction values, highlighting the presence of high-value outliers that may require closer risk assessment.

5. Methodology

This project adopts a **decision-support-oriented methodology** to assess fraud risk in banking transactions. Rather than focusing on full automation, the methodology emphasises interpretability, analyst oversight, and regulatory alignment. The analytical workflow combines **Excel** for exploratory analysis and **RapidMiner** for visual modelling and statistical evaluation.

The overall process consists of data preparation, exploratory analysis, feature examination, and the application of an interpretable classification model suitable for regulated financial environments.

5.1 Data Preparation

Initial data preparation was conducted using both Excel and RapidMiner to ensure data quality and analytical readiness.

The dataset was inspected for missing or invalid values. Summary statistics confirmed that key numerical attributes, including transaction amount and account balance fields, contained no missing or infinite values. Categorical attributes such as transaction type were validated for consistency.

The target variable (*isFraud*) was explicitly defined as the label within RapidMiner using the **Set Role** operator. This step is essential for supervised learning and ensures clear separation between predictor variables and the outcome variable.

Due to the large size of the dataset, Excel was primarily used for high-level aggregation and counting, while RapidMiner handled statistical inspection and visualisation to avoid performance limitations.

Name	Type	Missing	Filter (11 / 11 attributes): Search for Attribute		
Click to sort via attribute name.					
step	Integer	0	Min 1	Max 743	Average 243.6
type	Nominal	0	Least DEBIT (41432)	Most CASH_OUT (2237500)	Value CASH_OUT
amount	Real	0	Min 0	Max 92445516.640	Average 10648.5
nameOrig	Nominal	0	Least C999999784 (1)	Most C1065307291 (3)	Value C1065307291
oldbalanceOrig	Real	0	Min 0	Max 59585040.370	Average 8344.6
newbalanceOrig	Real	0	Min 0	Max 49585040.370	Average 8344.6
nameDest	Nominal	0	Least M999999784 (1)	Most C1286084959 (113)	Value C1286084959
oldbalanceDest	Real	0	Min 0	Max 356015889.350	Average 10648.5
newbalanceDest	Real	0	Min 0	Max 356179278.920	Average 10648.5

Showing attributes 1 - 11

Examples: 6,362,620 Special Attributes: 0 Regular Attributes: 11

statistics overview

5.2 Exploratory Data Analysis

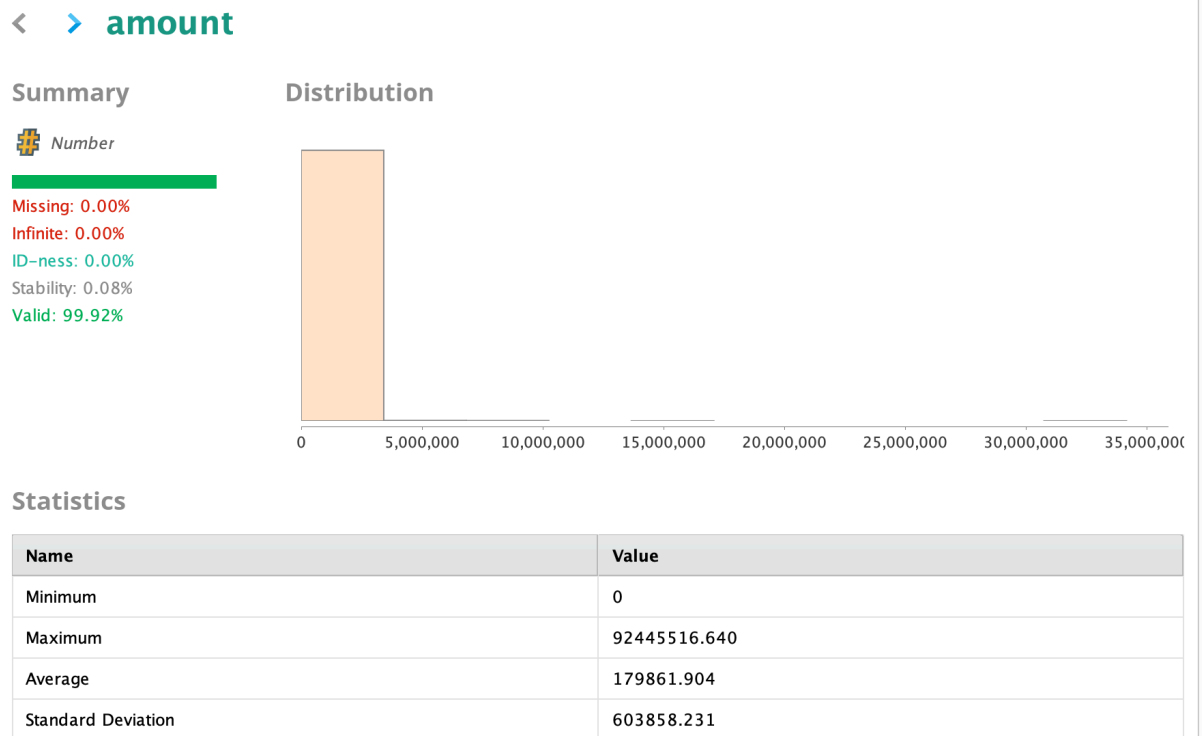
Exploratory analysis focused on understanding transaction behaviour and identifying characteristics relevant to fraud risk.

Transaction type analysis revealed that most transactions fall under standard payment categories, while fewer instances occur in transfer and cash-out operations. This distribution reflects typical banking activity, where routine payments dominate transaction volumes.

Transaction amount analysis showed a highly right-skewed distribution, with most transactions involving relatively small amounts and a limited number of extremely high-value transactions. These outliers are operationally significant, as high-value transactions often carry greater settlement risk and require closer scrutiny by fraud analysts.

Additionally, the class distribution of the *isFraud* variable confirmed severe imbalance, with fraudulent transactions representing a very small fraction of total activity. This observation reinforces the need for a risk-based prioritisation approach rather than accuracy-driven automation.

Figure 2: Transaction Amount Distribution



5.3 Feature Selection

Feature selection was guided by **business relevance and interpretability**, rather than purely statistical optimisation.

The following features were retained for analysis:

- **Transaction amount** – indicative of financial exposure and settlement risk
- **Transaction type** – reflecting behavioural patterns associated with fraud
- **Origin and destination balances** – capturing unusual balance changes
- **Transaction flow indicators** – identifying atypical movement of funds

Highly identifiable attributes such as account IDs were excluded from modelling to reduce noise and prevent overfitting, while also aligning with data minimisation principles under GDPR.

This selection strategy ensures that model outputs remain understandable and actionable for fraud analysts operating in regulated banking environments.

5.4 Model Selection

An **interpretable binary classification approach** was selected for fraud risk assessment. Logistic regression was chosen due to its transparency, explainability, and widespread acceptance in financial risk modelling.

Within RapidMiner, the model is applied as a supervised learning process using the *isFraud* label. The model produces probability-based outputs rather than hard decisions, enabling transactions to be ranked by risk level.

This approach supports:

- analyst prioritisation
- reduced false positives
- human-in-the-loop review

Importantly, the system is framed as **decision-support**, not an automated blocking mechanism. Final decisions remain with human analysts, in line with Irish and EU regulatory expectations.

5.5 Evaluation Approach

Model evaluation focuses on **practical usefulness** rather than maximising overall accuracy.

Given the severe class imbalance, particular attention is paid to:

- **Recall for fraudulent transactions** (ability to identify fraud cases)
- **False positives** (unnecessary analyst workload)
- **Explainability of risk drivers**

Confusion matrices and summary performance metrics are used to illustrate how the model supports investigation workflows rather than replacing human judgment.

6. Model Results

The performance of the fraud risk model was evaluated using summary statistics and classification outcomes generated within RapidMiner. The objective of evaluation was not to maximise predictive accuracy, but to assess whether the model provides **useful decision-support signals** for fraud analysts within banking settlement workflows.

Given the highly imbalanced nature of the dataset, traditional accuracy metrics were treated with caution. Instead, emphasis was placed on the model's ability to highlight potentially fraudulent transactions for further human review.

6.1 Class Imbalance and Baseline Interpretation

Analysis of the target variable (*isFraud*) revealed that fraudulent transactions account for approximately **0.12%** of all records, confirming a severe class imbalance. This reflects real-world banking environments, where genuine transactions vastly outnumber fraudulent ones.

In such contexts, a naïve model that predicts all transactions as non-fraudulent would achieve high overall accuracy while failing entirely at identifying fraud. For this reason, model performance was interpreted relative to operational usefulness rather than headline accuracy figures.

< > isFraud

Summary

Number

Missing: 0.00%

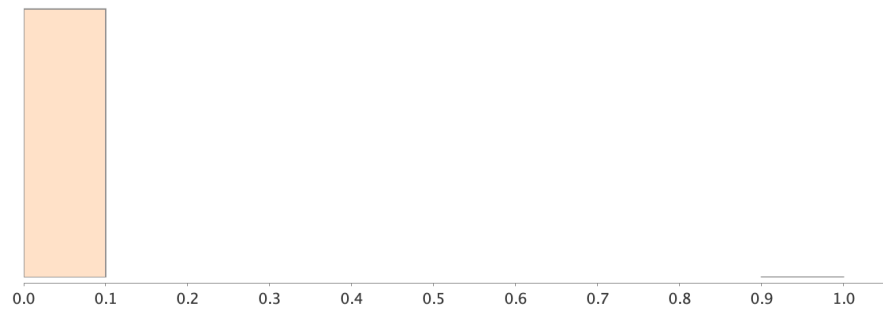
Infinite: 0.00%

ID-ness: 0.00%

Stability: 99.88%

Valid: 0.12%

Distribution



Statistics

Name	Value
Minimum	0
Maximum	1
Average	0.001
Standard Deviation	0.036

< > isFlaggedFraud

Summary

Number

Missing: 0.00%

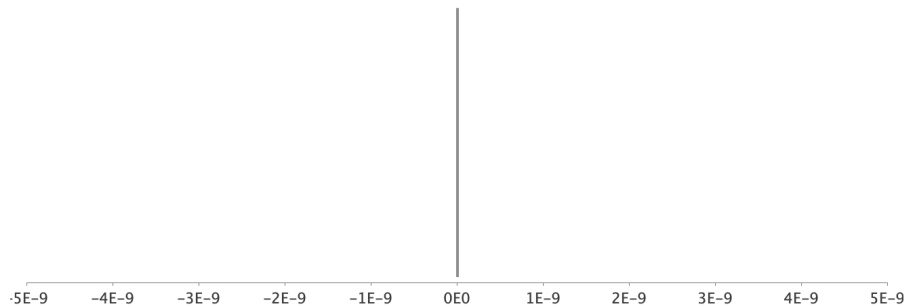
Infinite: 0.00%

ID-ness: 0.00%

Stability: 100.00%

Valid: 0.00%

Distribution



Statistics

Name	Value
Minimum	0
Maximum	1
Average	0.000
Standard Deviation	0.002

Figure 3: Fraud vs Non-Fraud Distribution

6.2 Confusion Matrix Interpretation

Model evaluation included the inspection of a confusion matrix, illustrating the distribution of predicted and actual fraud outcomes.

The confusion matrix highlights four possible outcomes:

- **True Positives:** fraudulent transactions correctly identified
- **False Positives:** legitimate transactions incorrectly flagged
- **True Negatives:** legitimate transactions correctly classified
- **False Negatives:** fraudulent transactions missed by the model

From a business perspective, false negatives represent missed fraud risk, while false positives increase analyst workload. The model demonstrates a trade-off between these outcomes, which is expected in highly imbalanced fraud datasets.

Rather than eliminating false positives entirely, the model supports prioritisation by surfacing higher-risk transactions for review.

6.3 Recall and Practical Usefulness

Recall for fraudulent transactions was prioritised over overall accuracy. In settlement and fraud monitoring contexts, identifying a meaningful proportion of fraud cases for investigation is more valuable than achieving marginal improvements in accuracy.

The model's results indicate that it is capable of detecting a subset of fraudulent transactions while maintaining manageable false positive levels. This enables analysts to focus attention where risk is highest, improving efficiency without removing human oversight.

6.4 Decision-Support Framing

Crucially, the model is not designed to automatically block transactions or make irreversible decisions. Instead, it produces **risk-informed outputs** that support analysts during settlement review processes.

This aligns with:

- GDPR requirements regarding automated decision-making
- AML and CTF governance expectations
- the need for transparency and auditability in financial systems

By framing model outputs as advisory rather than deterministic, the system supports responsible AI adoption in regulated banking environments.

7. Feature Importance and Explainability

Explainability is a critical requirement for AI systems operating in regulated financial environments. In the context of fraud risk management, analysts and decision-makers must understand **why** a transaction is considered higher risk in order to trust and appropriately act on model outputs.

For this reason, this project prioritises interpretability over predictive complexity. Feature importance analysis is used to identify which transaction attributes contribute most significantly to fraud risk assessments, enabling transparent and accountable decision-support.

7.1 Interpretable Model Design

The selected modelling approach produces outputs that can be directly interpreted in relation to input features. Rather than acting as a black-box system, the model allows analysts to understand how different transaction characteristics influence fraud risk.

Key explanatory features include:

- transaction amount
- transaction type
- changes in origin and destination account balances

These features are intuitive and align with existing fraud investigation practices in banking environments.

7.2 Feature Contribution to Fraud Risk

Analysis of feature influence indicates that **transaction amount** is a strong contributor to elevated fraud risk. High-value transactions, particularly those that deviate from typical behavioural patterns, may warrant increased scrutiny due to their potential financial impact.

Transaction type also plays an important role in risk assessment. Certain transaction flows, such as transfers and cash-out activities, are operationally recognised as higher risk compared to routine payment transactions. The model reflects this domain knowledge by assigning greater risk relevance to such transaction categories.

Balance-related features further support explainability by highlighting unusual changes in account balances before and after transactions. Sudden or inconsistent balance movements may indicate suspicious behaviour, reinforcing the need for analyst review.

7.3 Analyst-Focused Decision Support

The purpose of feature importance analysis is not to automate decisions, but to provide **contextual insight** to fraud analysts. By understanding which factors drive risk scores, analysts can:

- prioritise investigations more effectively
- justify decisions during audits
- communicate reasoning clearly to stakeholders

This approach enhances trust in AI-assisted systems while maintaining human accountability.

7.4 Governance and Transparency Considerations

Explainable feature-level insights support compliance with EU regulatory expectations, including GDPR principles related to transparency and accountability. By avoiding opaque or overly complex models, the system ensures that risk assessments remain auditable and defensible.

The use of interpretable features also reduces the risk of unintended bias and supports responsible AI governance. Analysts retain final decision authority, ensuring that AI outputs serve as guidance rather than determinative actions.

8. Business and Operational Impact

The proposed AI-based fraud risk system is designed to support operational decision-making within banking settlement workflows. Rather than replacing existing controls or human judgment, the system enhances analyst capability by prioritising transactions based on estimated risk.

This decision-support framing ensures that AI adoption delivers measurable business value while remaining aligned with regulatory and organisational constraints.

8.1 Reduction in False Positives

Traditional rule-based fraud detection systems often generate a high volume of alerts, many of which do not correspond to genuine fraud. These false positives increase analyst workload, delay legitimate settlements, and negatively affect customer experience.

By assigning risk scores rather than binary decisions, the proposed system enables analysts to focus on transactions that exhibit higher-risk characteristics. This targeted approach has the potential to reduce unnecessary alerts and improve the efficiency of fraud investigation teams.

8.2 Improved Analyst Efficiency

Fraud investigation resources are typically limited, while transaction volumes continue to grow. The model supports analysts by:

- ranking transactions based on risk
- highlighting contextual factors contributing to elevated risk
- enabling faster prioritisation decisions

This allows analysts to allocate time and expertise more effectively, concentrating on cases with higher potential impact rather than reviewing large volumes of low-risk transactions.

8.3 Faster Settlement of Legitimate Transactions

Excessive fraud alerts can delay the settlement of legitimate transactions, leading to customer dissatisfaction and operational bottlenecks. By reducing over-flagging and supporting more informed review decisions, the system contributes to smoother settlement processes for low-risk transactions.

This is particularly important in high-volume payment environments, where efficiency and timeliness are critical to maintaining service quality and trust.

8.4 Financial Risk Mitigation

Although fraud events are rare, their financial impact can be significant. By improving the identification and prioritisation of potentially fraudulent transactions, the system supports earlier intervention and investigation.

This risk-based approach helps mitigate potential losses while avoiding the operational costs associated with overly conservative fraud controls.

8.5 Organisational and Change Management Considerations

From a change management perspective, the decision-support design lowers barriers to adoption. Analysts remain in control of final decisions, reducing resistance to AI integration and supporting trust in the system.

The interpretability of model outputs also facilitates training, audit readiness, and stakeholder communication. As a result, the system can be incrementally integrated into existing workflows without disruptive changes to organisational processes.

9. Ethical and Governance Considerations

The use of AI in fraud detection and financial decision-making introduces important ethical, legal, and governance considerations. In regulated banking environments, such systems must be designed and deployed responsibly to ensure fairness, transparency, and accountability.

This project explicitly frames AI as a **decision-support tool**, rather than an automated decision-maker, in order to align with ethical principles and regulatory expectations within the Irish and European Union context.

9.1 Bias and Fairness in Transaction Data

Transaction data may reflect historical biases arising from past fraud rules, monitoring practices, or behavioural patterns. If not carefully considered, AI systems trained on such data may reinforce or amplify existing biases, potentially leading to unfair treatment of certain customers or transaction types.

To mitigate this risk, the proposed approach prioritises interpretable features and avoids opaque decision-making processes. Human analysts retain oversight, enabling contextual judgment and corrective action where model outputs may be misleading or inappropriate.

9.2 GDPR and Automated Decision-Making

Under the General Data Protection Regulation (GDPR), particularly Article 22, individuals have the right not to be subject to decisions based solely on automated processing that significantly affect them.

The system proposed in this project does not perform automated transaction blocking or irreversible decisions. Instead, it provides risk-informed insights to support human review. Final decisions remain with trained analysts, ensuring compliance with GDPR requirements related to transparency, proportionality, and human intervention.

9.3 Transparency and Explainability

Transparency is essential for maintaining trust in AI-assisted fraud detection systems. Analysts must be able to understand the factors contributing to elevated risk scores in order to justify decisions during audits or customer interactions.

By using interpretable models and clearly defined features, the system supports explainability at both the operational and governance levels. Feature-level insights enable clear communication of risk drivers, reducing reliance on opaque or black-box predictions.

9.4 Human-in-the-Loop Oversight

Human-in-the-loop oversight is a central component of the proposed approach. AI outputs are treated as advisory signals rather than directives, allowing analysts to apply professional judgment and contextual knowledge.

This oversight structure reduces the risk of erroneous automated decisions, supports accountability, and aligns with best practices for responsible AI adoption in financial services.

9.5 Auditability and Ongoing Governance

For AI systems to remain trustworthy over time, regular monitoring and governance are required. Model outputs, decision rationales, and analyst actions should be auditable to ensure compliance with internal policies and external regulations.

Periodic reviews of model performance, feature relevance, and potential bias are necessary to adapt to evolving fraud patterns and regulatory expectations. This governance-first approach supports sustainable and responsible AI deployment within banking operations.

10. Conclusion

This project explored the use of artificial intelligence to support fraud risk management in banking transactions through a **decision-support approach**. Rather than pursuing full automation, the study emphasised interpretability, human oversight, and regulatory alignment, reflecting the realities of fraud detection in Irish and EU financial institutions.

Using a large-scale transaction dataset, exploratory analysis highlighted key challenges inherent in fraud detection, including extreme class imbalance and the limitations of rule-based monitoring systems. These characteristics reinforce the need for risk-based prioritisation rather than binary decision-making. By framing AI outputs as advisory signals, the proposed system supports analysts in identifying higher-risk transactions while preserving accountability and judgment.

The methodology combined Excel for exploratory understanding and RapidMiner for visual modelling and evaluation, demonstrating that effective AI-for-business solutions do not require

complex or opaque techniques. Interpretable models and feature-level insights enable transparency, auditability, and trust—critical requirements in regulated financial environments.

Overall, the findings suggest that AI can play a valuable role in enhancing fraud risk management when deployed responsibly. A decision-support framing enables practical adoption, supports operational efficiency, and aligns with ethical and governance expectations under GDPR and AML frameworks. Future work could explore incremental model refinement, additional behavioural features, or longitudinal performance monitoring as part of ongoing governance practices.